Network Science, Week 2

The Zachary Karate Club network represents the social relationships among 34 members of a university karate club observed in the 1970s. Nodes correspond to club members, and edges indicate friendships outside of club activities. The network split into factions after a conflict between the instructor and the administrator, making it a classic example for studying community structure and network division.

Part 1: Getting Started

1. Download and import the Zachary Karate Club network (CSV edge list).
    - How many nodes and edges are in the network?
    - Identify the node with the highest degree.
2. Explore the Network Panel: toggle **Show Number of Nodes and Edges**.

Part 2: Network Visualization

3. Apply different layouts: Force-directed, Circular, Hierarchical.
    - Which layout best shows the natural communities in the network?
4. Change node sizes based on **degree centrality**.
    - Which nodes are the most connected?
5. Map the Edge Width or Edge Color in the Style to the EdgeBetweenness column using a Continuous Mapping.
    - Are there any critical connections in terms of the overall network's connectivity?

Part 3: Community Detection

6. Use ClusterMaker2 or Cytoscape community detection to detect clusters.
    - How many clusters are detected?
    - Do they match the known division in the Karate Club?
7. Color nodes by cluster.

Part 4: Graph Analysis

8. Calculate network metrics:
    - Average degree
    - Density
    - Diameter
    - Betweenness centrality

- Which node is most critical for connectivity?

9. Filter nodes with degree < 2 and observe changes.

Part 5: Reflection

10. Compare layouts and metrics:
    - How does network visualization help interpret node importance and clusters?
    - How could these techniques apply to social network or recommendation system datasets?
    - Save .cys sessions to track your network manipulations.

Summary of metrics (Node Table + Edge Table):

Metric	Definition	How to Interpret
Degree	Number of direct neighbors (edges).	High = local hub, well-connected.
Betweenness Centrality	Fraction of shortest paths that pass through the node. Measures bridging/brokerage role.	High = key connector between groups.
Closeness Centrality	Inverse of average shortest path length to all other nodes.	High = node is “close” to everyone, spreads info quickly.
Radiality	Centrality measure scaled by graph diameter. Combines closeness with normalization.	High = very central; low = peripheral.
Clustering Coefficient	Fraction of a node’s neighbors that are also neighbors of each other (triangles).	High = belongs to a tight-knit clique.
Topological Coefficient	Similarity of a node’s neighbors compared to its neighbors’ neighborhoods.	High = node shares many common neighbors with its neighbors. Low = node connects otherwise unconnected groups.
Neighborhood Connectivity	Average degree of a node’s neighbors.	High = connects mainly to other hubs; low = connects to low-degree nodes.
Edge Betweenness	Shows edge's importance as a bridge	High = critical to overall network connectivity